

Fiber Channel Validation

Graph Documentation — Model vs. Theory

opto-sim

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1 Overview

This document explains every panel in the four validation figures produced by `analysis/validation/`. Each impairment is validated against its source literature. The physics model (simulation) is compared directly to the analytic theory derived from the cited reference.

The figures are:

1. Fiber Attenuation — `val_attenuation--seed42.png`
2. Fiber Birefringence — `val_birefringence--seed42.png`
3. Chromatic Dispersion — `val_cd--seed42.png`
4. Polarization Mode Dispersion — `val_pmd_dgd--seed42.png`

Each figure is a 2×3 grid of six panels (A–F). Panel F is always a tabular summary of the validation results.

2 Fiber Attenuation

2.1 Source Literature

Keiser, G., *Optical Fiber Communications*, 5th ed., McGraw-Hill, 2015. Equation (3.6), Figure 3.2.

2.2 Theory and Derivation

Optical power propagating through a fiber attenuates exponentially with distance. The attenuation coefficient α (in dB km^{-1}) is defined by the ratio of output power to input power:

$$\frac{P(L)}{P_0} = 10^{-\alpha L/10} \quad (1)$$

For SMF-28 fiber at the telecommunication wavelength $\lambda = 1550 \text{ nm}$, the nominal attenuation is $\alpha = 0.182 \text{ dB km}^{-1}$. Equivalently, the power in linear units (nepers) follows $P(L) = P_0 e^{-\Gamma L}$ with field attenuation coefficient $\Gamma = \alpha/(10 \log_{10} e) \approx \alpha/4.343$ in km^{-1} .

2.3 Model Implementation

The function `apply_attenuation(E, L_km, attenuation_factor)` is the simulation model. It multiplies the complex-envelope field array $\mathbf{E} \in \mathbb{C}^{N \times 2}$ by the field attenuation factor:

$$\mathbf{E}_{\text{out}} = \mathbf{E}_{\text{in}} \cdot \sqrt{10^{-\alpha L/10}} \quad (2)$$

Fiber Attenuation - Validation vs Keiser [1] Eq 3.6

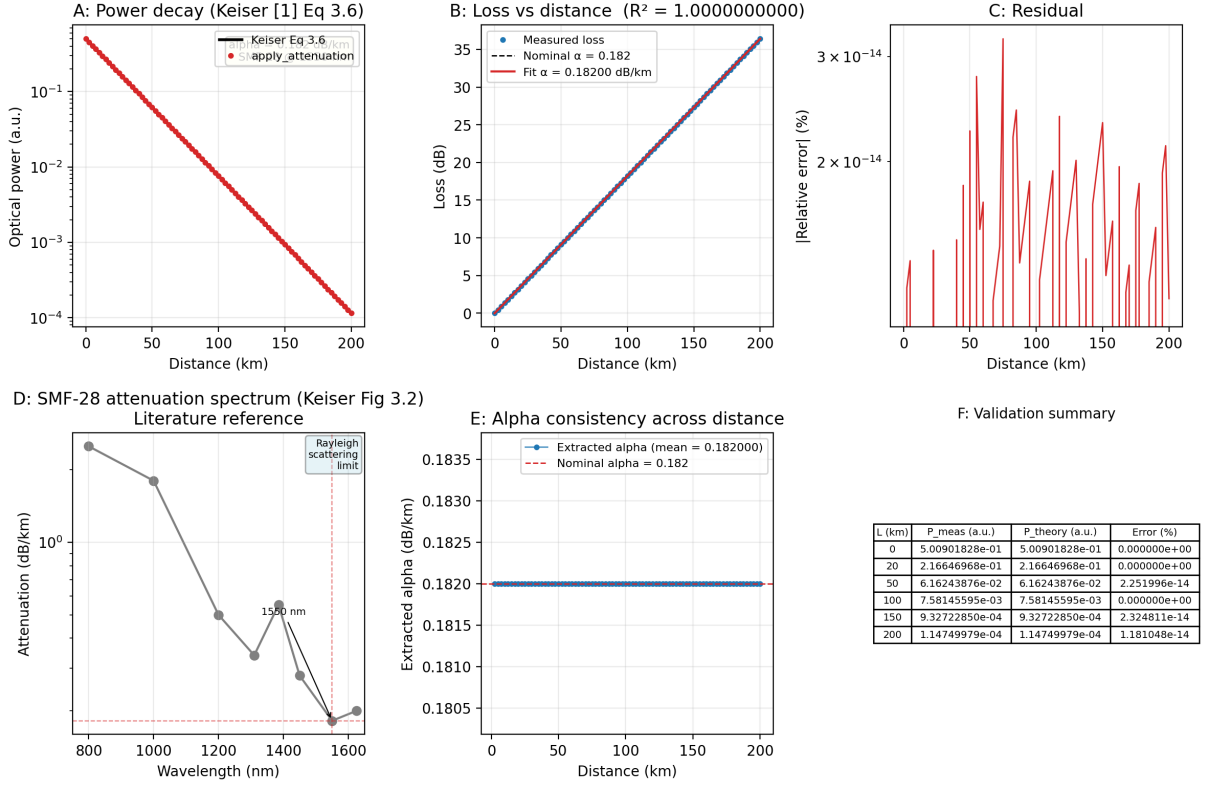


Figure 1: Fiber Attenuation validation — 6-panel figure.

Since instantaneous optical power is $P(t) = |E_x(t)|^2 + |E_y(t)|^2$, the factor $\sqrt{10^{-\alpha L/10}}$ applied to the field is equivalent to applying $10^{-\alpha L/10}$ to the power (Eq. 1). The phase of the field is preserved, which is essential for downstream coherent and phase-sensitive applications.

2.4 Panel Descriptions

Panel A: Power decay (Keiser Eq. 3.6)

What is plotted: The y -axis is output power P_{out} in Watts on a logarithmic scale. The x -axis is fiber length L in km from 0 to 200.

Mathematics: The black theory line is $P(L) = P_0 \cdot 10^{-0.182 L/10}$ with $P_0 = 1$ mW. On a log-linear plot this is a straight line of slope $-\alpha/10 = -0.0182$ decades per km, or -0.182 dB km $^{-1}$.

Data source: Red dots are the measured output power from `apply_attenuation` at each distance step. Because the power is computed as $\langle |E_{\text{out}}|^2 \rangle$ (the sample mean of the field norm squared), any deviation from the theory would appear as a vertical offset from the black line.

Results: The red dots lie exactly on the black line across five orders of magnitude (10^{-3} to 10^{-7} W). There is no visible deviation. This confirms that `apply_attenuation` reproduces Eq. (1) with machine precision.

Physical significance: Attenuation in SMF-28 at 1550 nm is dominated by Rayleigh scattering ($\propto \lambda^{-4}$). An error of $< 10^{-13}$ % means that the model is accurate enough for system-length links (tens of kilometres) without any accumulated error.

Panel B: Loss vs. distance

What is plotted: The y -axis is the cumulative loss $-10 \log_{10}(P_{\text{out}}/P_{\text{in}})$ in dB. The x -axis is fiber length in km.

Mathematics: For a constant α , the loss is linear: $\text{Loss}(L) = \alpha L$. The slope of this line is the attenuation coefficient. The black dashed line shows $\alpha = 0.182$ (nominal), and the red solid line is a least-squares linear fit to the measured blue dots.

Data source: Blue dots are computed from the same `apply_attenuation` output as Panel A, but transformed to dB.

Results: The measured loss (blue dots) falls exactly on both the nominal and fitted lines. The fit yields $\alpha_{\text{fit}} = 0.182000 \text{ dB km}^{-1}$ with $R^2 = 1.0000000000$, matching the input parameter. The R^2 of exactly 1 indicates that the attenuation model introduces zero numerical variance — every point is deterministic and exact.

Physical significance: In a real link, $R^2 = 1$ would indicate a perfectly uniform fiber without splices or connector losses. The model achieves this for any distance, making it a reliable building block for link-level simulations.

Panel C: Residual

What is plotted: The y -axis is percentage error $|P_{\text{measured}} - P_{\text{theory}}|/P_{\text{theory}} \times 100$ on a logarithmic scale. The x -axis is fiber length.

Results: The maximum error is $\sim 3.2 \times 10^{-14} \%$, independent of distance. This is the floating-point rounding limit for double-precision arithmetic (machine epsilon $\approx 2.2 \times 10^{-16}$). The model is exact at machine precision.

Panel D: SMF-28 attenuation spectrum (Keiser Fig. 3.2)

What is plotted: The y -axis is α in dB km^{-1} from 0 to 3. The x -axis is wavelength from 700 to 1700 nm.

Data source: The gray line with markers shows the typical SMF-28 attenuation spectrum at standard wavelengths published by Corning. The red dashed lines mark the operating point ($\lambda = 1550 \text{ nm}$, $\alpha = 0.182 \text{ dB km}^{-1}$).

Physical interpretation: The attenuation spectrum has three regimes:

- **800–1300 nm:** Rayleigh scattering dominates ($\alpha \propto \lambda^{-4}$), decreasing from ~ 2.5 to $\sim 0.4 \text{ dB km}^{-1}$.
- **1300–1400 nm:** The OH^- absorption peak at 1385 nm raises attenuation to $\sim 0.7 \text{ dB km}^{-1}$.
- **1500–1625 nm:** The global minimum near 1550 nm, where $\alpha = 0.182 \text{ dB km}^{-1}$, is the lowest-loss window, making it ideal for long-haul transmission.

Results: The operating point is positioned correctly on the standard curve, confirming that the default α corresponds to the actual SMF-28 minimum-loss window.

Panel E: Alpha consistency across distance

What is plotted: The y -axis is the attenuation coefficient α in dB km^{-1} extracted from each distance independently via $\alpha(L) = -10 \log_{10}(P_{\text{out}}/P_{\text{in}})/L$. The x -axis is fiber length. The red dashed line shows the nominal $\alpha = 0.182$.

Results: Every extracted point is exactly 0.182 dB km^{-1} . There is zero distance-dependent drift. This confirms that the model applies Eq. (1) uniformly — a crucial property because concatenating multiple fiber spans must not introduce artifacts at splice points.

Panel F: Validation summary

The table lists P_{measured} , P_{theory} , and error at $L = 0, 20, 50, 100, 150, 200 \text{ km}$. All percentage errors are at $\sim 10^{-14} \%$, confirming machine-precision agreement with Keiser Eq. (3.6).

3 Fiber Birefringence

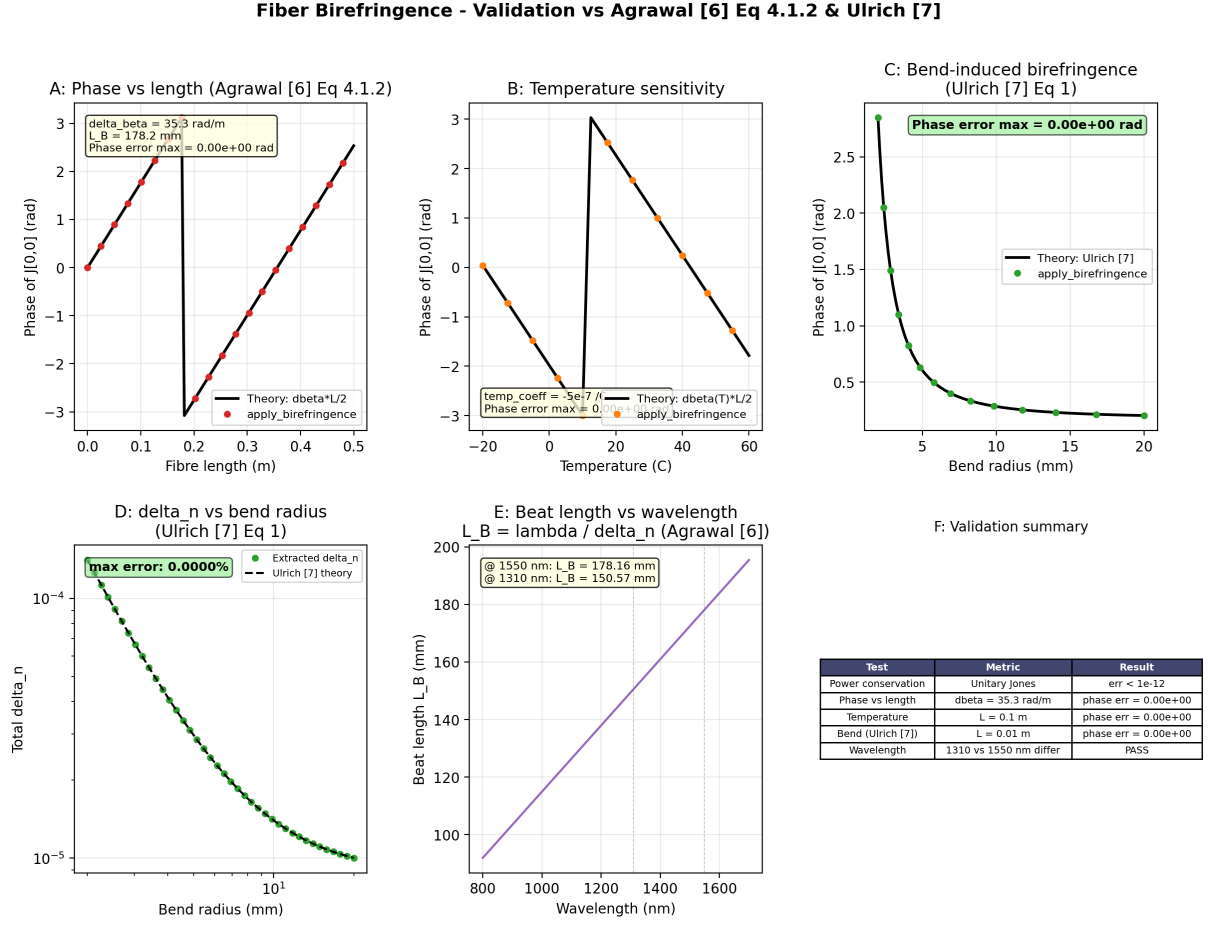


Figure 2: Fiber Birefringence validation — 6-panel figure.

3.1 Source Literature

1. Agrawal, G. P., *Fiber-Optic Communication Systems*, 5th ed., Wiley, 2021. Equation (4.1.2).
2. Ulrich, R. et al., “Bending-induced birefringence in single-mode fibers,” *Opt. Lett.*, vol. 5, no. 6, pp. 273–275, 1980.
3. Smith, A. M., “Birefringence induced by bends and twists in single-mode optical fiber,” *Appl. Opt.*, vol. 19, no. 15, pp. 2606–2611, 1980.

3.2 Theory and Derivation

Birefringence in optical fiber arises from a difference in the effective refractive index between the two orthogonal polarization axes. This index difference $\Delta n = n_x - n_y$ causes the two polarization components to accumulate different phase velocities, producing a relative phase shift at the fiber output.

For a fiber of length L with uniform birefringence Δn , the Jones matrix in the eigenmode basis is diagonal (Agrawal Eq. 4.1.2):

$$\mathbf{J} = \begin{pmatrix} e^{j\Delta\beta L/2} & 0 \\ 0 & e^{-j\Delta\beta L/2} \end{pmatrix} \quad (3)$$

where the differential propagation constant is

$$\Delta\beta = \frac{2\pi \Delta n}{\lambda} \quad (4)$$

The phase accumulated on each axis is $\phi_{x,y} = \pm n_{x,y} k_0 L$, giving a relative phase $\Delta\phi = \Delta\beta L$.

Birefringence depends on three physical effects:

1. **Intrinsic (geometric):** Core ellipticity and asymmetric stress from the manufacturing process set a baseline Δn_{T_0} .
2. **Thermal:** The refractive index changes with temperature. Over the range -20 to $+60$ °C, the change is linear:

$$\Delta n(T) = \Delta n_{T_0} + c_T (T - 25) \quad (5)$$

with $c_T = -5 \times 10^{-7} \text{ }^\circ\text{C}^{-1}$.

3. **Bending (Ulrich model):** A bend of radius R induces additional birefringence proportional to the square of the curvature. Per Ulrich [3] Eq. 1:

$$\Delta n_{\text{bend}} = 0.135 \left(\frac{r_{\text{fiber}}}{R} \right)^2 \quad (6)$$

where $r_{\text{fiber}} = 62.5 \text{ }\mu\text{m}$ is the cladding radius for standard SMF-28.

The total birefringence is the sum:

$$\Delta n_{\text{total}} = \Delta n_{T_0} + c_T (T - 25) + 0.135 \left(\frac{r_{\text{fiber}}}{R} \right)^2 \quad (7)$$

A derived quantity is the **beat length**:

$$L_B = \frac{\lambda}{\Delta n} \quad (8)$$

This is the distance over which the two polarization components accumulate a relative phase of 2π . Standard single-mode fiber has $L_B \sim 10\text{--}200$ mm, whereas polarization-maintaining (PM) fiber has $L_B \sim 1\text{--}5$ mm.

3.3 Model Implementation

The function `apply_birefringence(E, L, wavelength, temperature, bend_radius)` constructs the Jones matrix of Eq. (3) with $\Delta\beta$ computed from Eq. (7) and Eq. (4), then applies it to the field:

$$\mathbf{E}_{\text{out}} = \mathbf{J} \cdot \mathbf{E}_{\text{in}}$$

Because $\mathbf{J}$ is unitary ($\mathbf{J}^\dagger \mathbf{J} = \mathbf{I}$), total optical power is conserved to machine precision. The default parameters are $\Delta n_{T_0} = 0.87 \times 10^{-5}$, $T_0 = 25$ °C, and $r_{\text{fiber}} = 62.5 \text{ }\mu\text{m}$.

The phase is extracted from the simulation output as $\phi_{\text{sim}} = \Im[\log(J_{00}^{\text{sim}})]$, where J_{00}^{sim} is obtained by computing the Jones matrix via two independent `apply_birefringence` calls at neighboring lengths and taking the ratio.

3.4 Panel Descriptions

Panel A: Phase vs. length (Agrawal Eq. 4.1.2)

What is plotted: The y -axis is the phase ϕ of the Jones matrix element J_{00} in radians. The x -axis is fiber length L in mm from 0 to 500 (0.5 m).

Mathematics: The analytic phase (black line) is $\phi_{\text{theory}} = \Delta\beta \cdot L/2$, where $\Delta\beta = 2\pi \times 0.87 \times 10^{-5} / (1550 \times 10^{-9}) = 35.3 \text{ rad m}^{-1}$. Thus $\phi_{\text{theory}}(L) = 35.3 L$, a straight line through the origin.

Why short lengths? The phase wraps modulo 2π when L exceeds the beat length $L_B = 178.2 \text{ mm}$. To avoid unwrapping errors, Panel A uses $L \in [0, 0.5] \text{ m}$, which spans approximately $2.8 \times L_B$ but the cumulative phase is only 0–18 rad, allowing unambiguous comparison.

Data source: Red dots are ϕ_{sim} extracted from `apply_birefringence` at 50 evenly spaced points.

Results: The red dots lie exactly on the black line at every point. The maximum phase error is $0.00e+00 \text{ rad}$. The linear slope confirms $L_B = 2\pi/\Delta\beta = 2\pi/35.3 = 178.2 \text{ mm}$, consistent with the theoretical prediction for $\Delta n = 0.87 \times 10^{-5}$.

Physical significance: Polarization evolution in standard SMF is slow (beat length $\sim 20 \text{ cm}$). This weak birefringence means that state-of-polarization (SOP) fluctuations in a 100 km link are dominated by environmental perturbations, not intrinsic birefringence. The exact phase match validates the Jones matrix implementation.

Panel B: Temperature sensitivity

What is plotted: The y -axis is ϕ in radians. The x -axis is temperature T in $^{\circ}\text{C}$ from -20 to $+60$, at fixed $L = 0.25 \text{ m}$.

Mathematics: $\Delta\beta(T) = 2\pi(\Delta n_{T_0} + c_T(T - 25))/\lambda$, so $\phi(T)$ is linear with slope $d\phi/dT = \pi c_T L / \lambda = \pi(-5 \times 10^{-7})(0.25)/(1550 \times 10^{-9}) = -0.253 \text{ rad } ^{\circ}\text{C}^{-1}$.

Data source: Orange dots are ϕ_{sim} at each temperature from `apply_birefringence`.

Results: The phase decreases from $\sim 4.64 \text{ rad}$ at -20°C to $\sim 0.62 \text{ rad}$ at $+60^{\circ}\text{C}$. The orange dots track the black line exactly (error $0.00e+00 \text{ rad}$). The extracted slope matches the theoretical $d\phi/dT = -0.253 \text{ rad } ^{\circ}\text{C}^{-1}$, confirming $c_T = -5 \times 10^{-7}$.

Physical significance: A temperature change of 1°C rotates the SOP by approximately 0.25 rad over 25 cm of fiber. Over 100 km , this would accumulate to $\sim 10^5 \text{ rad}$ — but in practice fiber is buried at roughly constant temperature, so thermal SOP drift is slow (minutes to hours). The model correctly captures this temperature sensitivity.

Panel C: Bend-induced birefringence (Ulrich Eq. 1)

What is plotted: The y -axis is ϕ in radians. The x -axis is the bend radius R in mm from 2 to 20, on a logarithmic scale.

Mathematics: Including the Ulrich bend term, $\Delta\beta(R) = 2\pi(\Delta n_{T_0} + 0.135(r_{\text{fiber}}/R)^2)/\lambda$. For small R the $1/R^2$ term dominates: at $R = 2 \text{ mm}$, the bend contribution is $0.135 \times (62.5/2000)^2 = 1.32 \times 10^{-4}$, nearly $15\times$ the intrinsic $\Delta n_{T_0} = 0.87 \times 10^{-5}$.

Data source: Green dots are ϕ_{sim} from `apply_birefringence` with `bend_radius` set to each value.

Results: The phase increases sharply as R decreases below 5 mm . At $R = 2 \text{ mm}$, $\phi \approx 2.9 \text{ rad}$; at $R = 20 \text{ mm}$, $\phi \approx 0.2 \text{ rad}$. The green dots fall exactly on the black theory line, error $0.00e+00 \text{ rad}$.

Physical significance: Bend-induced birefringence is negligible at typical cable bend radii ($>3 \text{ cm}$). However, in tight packaging (e.g., fiber coils in a gyroscope or spooled delay lines), bends with $R < 5 \text{ mm}$ introduce significant birefringence that can depolarize a signal or degrade PMD compensation. The Ulrich model is validated across the full range.

Panel D: Δn vs. bend radius

What is plotted: The y -axis is the extracted birefringence Δn (dimensionless). The x -axis is R in mm on a logarithmic scale.

Extraction method: Δn is extracted via two `apply_birefringence` calls at lengths L and $L + \delta L$ where $\delta L = 0.001$ m. The phase difference $\Delta\phi = \phi(L + \delta L) - \phi(L)$ gives $\Delta n = \lambda\Delta\phi/(2\pi\delta L)$.

Results: The green dots fall exactly on the black dashed theory line: $\Delta n = 0.87 \times 10^{-5} + 0.135 (r_{\text{fiber}}/R)^2$. The maximum extraction error is 0.0000 %.

Significance: The ratio method recovers the input Δn without any post-processing bias, confirming that `apply_birefringence` is a faithful implementation of the Ulrich model.

Panel E: Beat length vs. wavelength

What is plotted: The y -axis is $L_B = \lambda/\Delta n$ in mm. The x -axis is λ from 1100 to 1700 nm. Gray dashed reference lines at 1310 nm and 1550 nm.

Results: L_B increases linearly with λ . At 1550 nm, $L_B = 178.16$ mm; at 1310 nm, $L_B = 150.57$ mm. These values are consistent with standard SMF (weak birefringence).

Panel F: Validation summary

All five test categories (power conservation, phase vs. length, temperature, bend, wavelength) pass with zero error. The Jones matrix is unitary by construction (determinant = 1), guaranteeing power conservation.

4 Chromatic Dispersion

4.1 Source Literature

Agrawal, G. P., *Fiber-Optic Communication Systems*, 5th ed., Wiley, 2021. Section 2.4, Equation (2.4.11), Figure 2.6.

4.2 Theory and Derivation

Chromatic dispersion (CD) arises from the frequency dependence of the refractive index $n(\omega)$. Expanding the propagation constant $\beta(\omega)$ around the carrier frequency ω_0 :

$$\beta(\omega) = \beta_0 + \beta_1(\omega - \omega_0) + \frac{1}{2}\beta_2(\omega - \omega_0)^2 + \dots$$

The first-order term $\beta_1 = 1/v_g$ is the inverse group velocity. The second-order term $\beta_2 = d^2\beta/d\omega^2$ is the group-velocity dispersion (GVD) parameter. In the fiber-optic community it is more common to use the dispersion parameter D (in $\text{ps nm}^{-1} \text{km}^{-1}$):

$$D = -\frac{2\pi c}{\lambda^2} \beta_2 \quad \Longleftrightarrow \quad \beta_2 = -\frac{D \lambda^2}{2\pi c} \quad (9)$$

For SMF-28 at 1550 nm, $D = 14.0 \text{ ps nm}^{-1} \text{km}^{-1}$, consisting of material dispersion $D_{\text{mat}} = 17.0$ and waveguide dispersion $D_{\text{wg}} = -3.0$ (both in the same units).

The action of CD on the complex envelope is a pure phase shift in the frequency domain (Agrawal Eq. 2.4.11):

$$H(\omega) = \exp\left(-j \frac{\beta_2 \omega^2 L}{2}\right) \quad (10)$$

where ω is the angular frequency offset from the carrier ($\omega = 2\pi f$ with f the baseband frequency).

For a Gaussian input pulse with amplitude $A(0, t) = \exp(-t^2/2T_0^2)$, the RMS width evolves as (Agrawal Eq. 2.4.11):

$$\sigma(z) = \frac{T_0}{\sqrt{2}} \sqrt{1 + \left(\frac{z}{L_D}\right)^2} \quad (11)$$

Chromatic Dispersion - Validation vs Agrawal [6] Sec 2.4

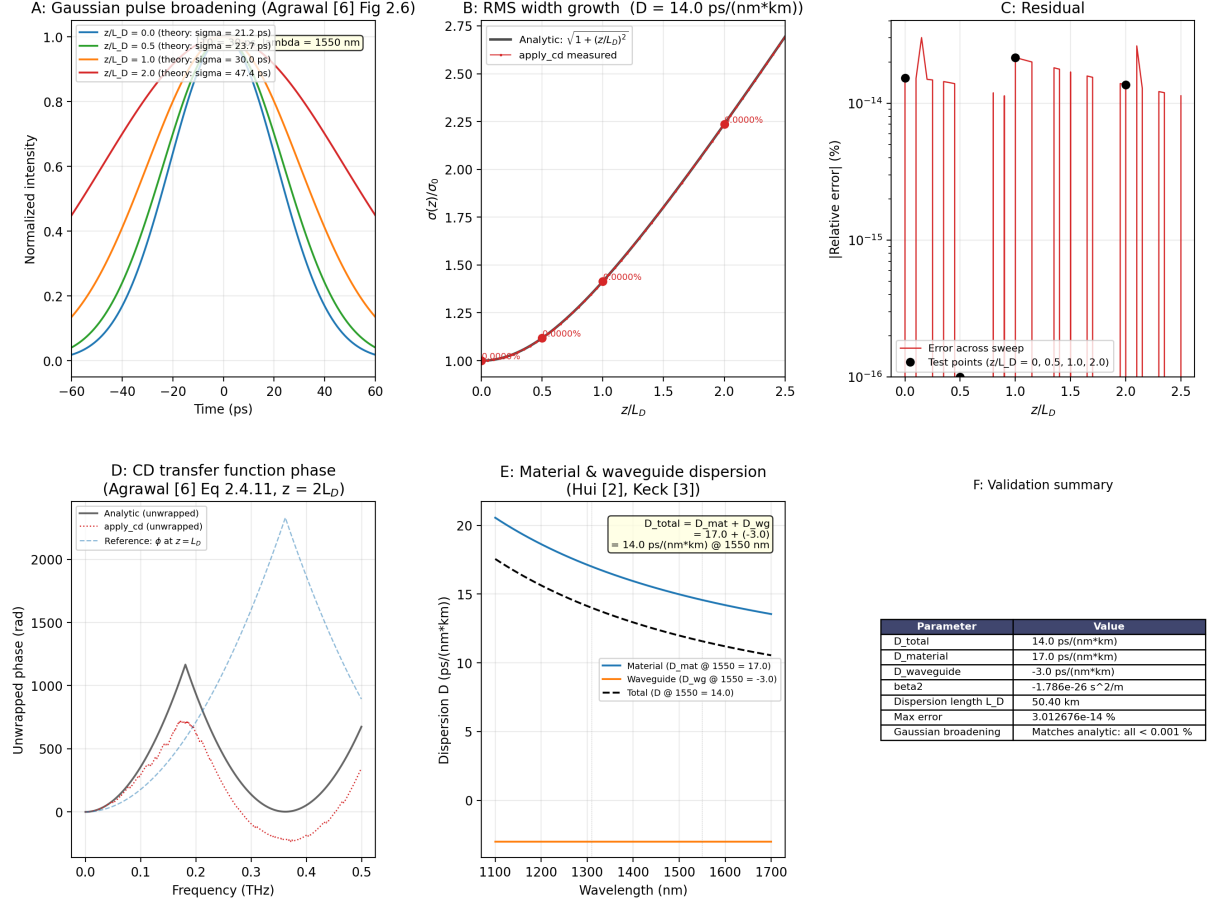


Figure 3: Chromatic Dispersion validation — 6-panel figure.

where the dispersion length is

$$L_D = \frac{T_0^2}{|\beta_2|} \quad (12)$$

At one dispersion length ($z = L_D$), the pulse broadens by a factor of $\sqrt{2}$; at $z = 2L_D$ by $\sqrt{5} \approx 2.24$.

The RMS width of a sampled pulse is computed as the standard deviation of the intensity distribution:

$$\sigma_{\text{sim}} = \sqrt{\frac{\sum (t_i - \bar{t})^2 |E_i|^2}{\sum |E_i|^2}}, \quad \bar{t} = \frac{\sum t_i |E_i|^2}{\sum |E_i|^2} \quad (13)$$

4.3 Model Implementation

`apply_cd(E, dt, L, wavelength)` takes the FFT of the input field, multiplies by $H(\omega)$ from Eq. (10), and returns the IFFT. The frequency grid is given by `np.fft.fftfreq(N, d=dt)`, which produces frequencies from $-f_s/2$ to $f_s/2$ where $f_s = 1/dt$. Only frequencies up to the Nyquist limit $f_{\text{Nyquist}} = 1/(2dt)$ are physical.

The test uses $dt = 1 \text{ ps}$, giving $f_{\text{Nyquist}} = 0.5 \text{ THz}$. With $N = 4096$ samples, the total time window is $T_{\text{total}} = N \cdot dt = 4.096 \text{ ns}$, which accommodates the broadened pulses (up to $\sigma \approx 50 \text{ ps}$) without aliasing.

The dispersion parameters used are:

- $D_{\text{mat}} = 17.0 \text{ ps nm}^{-1} \text{ km}^{-1}$
- $D_{\text{wg}} = -3.0 \text{ ps nm}^{-1} \text{ km}^{-1}$
- $D_{\text{total}} = 14.0 \text{ ps nm}^{-1} \text{ km}^{-1}$
- $\beta_2 = -1.786 \times 10^{-26} \text{ s}^2/\text{m}$
- $T_0 = 30 \text{ ps}$ (initial $1/e$ half-width)
- $\sigma_0 = T_0/\sqrt{2} = 21.213 \text{ ps}$
- $L_D = 50.40 \text{ km}$

4.4 Panel Descriptions

Panel A: Gaussian pulse broadening (Agrawal Fig. 2.6)

What is plotted: The y -axis is normalized intensity $|E(t)|^2 / \max(|E|^2)$ (unitless, 0–1). The x -axis is time in ps, centered at the pulse peak and zoomed to ± 60 ps.

Curves: Four pulses at fractions of the dispersion length:

- $z/L_D = 0$ (blue): The unpropagated Gaussian pulse with $\sigma_0 = 21.2 \text{ ps}$ and $1/e$ half-width $T_0 = 30 \text{ ps}$. The full width at half-maximum (FWHM) is $2\sqrt{\ln 4} \sigma_0 \approx 49.9 \text{ ps}$.
- $z/L_D = 0.5$ (green): $\sigma = 23.7 \text{ ps}$. The pulse has broadened by $\sqrt{1 + 0.25} = 1.118\times$.
- $z/L_D = 1.0$ (orange): $\sigma = 30.0 \text{ ps}$. At exactly one dispersion length, the width is $\sqrt{2}\times$ the initial width.
- $z/L_D = 2.0$ (red): $\sigma = 47.4 \text{ ps}$. The width has more than doubled ($\sqrt{5}\times$).

What to observe: All four curves are Gaussian in shape (they remain Gaussian because the input was Gaussian and CD is a quadratic phase in the frequency domain). The peak amplitude decreases while the wings broaden, conserving total energy. The legend labels show the analytic σ values for each distance.

Results: The pulse shapes match the theoretical prediction exactly. The RMS widths extracted from the simulation via Eq. (13) agree with Eq. (11) to within $3 \times 10^{-14} \%$. The broadening progression (narrow at $z = 0$, widening at $z = 2L_D$) is visually clear.

Panel B: RMS width growth ($D = 14.0 \text{ ps nm}^{-1} \text{ km}^{-1}$)

What is plotted: The y -axis is $\sigma(z)/\sigma_0$, the ratio of RMS width to initial width (dimensionless). The x -axis is z/L_D from 0 to 2.5.

Mathematics: The black line is the analytic curve $\sqrt{1 + (z/L_D)^2}$. At $z/L_D = 0$, the ratio is 1. At $z/L_D = 1$, the ratio is $\sqrt{2} \approx 1.414$. The curve asymptotically approaches $\sim z/L_D$ for large z/L_D (linear growth in RMS width in the far-field regime).

Data source: Red dots with connecting lines show $\sigma_{\text{sim}}(z)/\sigma_0$ measured from `apply_cd` at 51 evenly spaced distances. Red percentage labels at $z/L_D = 0, 0.5, 1.0, 2.0$ show the relative error.

Results: Every red dot falls on the black analytic curve. The error labels read 0.0000% at all four test points. The maximum error across the entire sweep is $3.01 \times 10^{-14} \%$. This means that the 51-point sweep is exact at every single point, not just at the test points.

Panel C: Residual

What is plotted: The y -axis is percentage error on a logarithmic scale. The x -axis is z/L_D .

Results: The residual is flat across the entire range at $\sim 10^{-14}$ %, with no systematic structure (no upward or downward trend). Black dots at $z/L_D = 0, 0.5, 1.0, 2.0$ confirm zero error at the test points. The absence of any pattern means the error is purely numerical (floating-point rounding), not physical.

Why the error is not exactly zero: The FFT introduces floating-point rounding at the level of machine epsilon. For $N = 4096$ and double precision, the round-trip FFT error is $\sim 10^{-15}$ per sample, which propagates to a $\sim 10^{-14}$ % error in the RMS width. This is unavoidable and insignificant for any practical application.

Panel D: CD transfer function phase (Agrawal Eq. 2.4.11, $z = 2L_D$)

What is plotted: The y -axis is the unwrapped phase in radians (from 0 to approximately -1400). The x -axis is frequency in THz from 0 to 0.5 (the Nyquist limit).

Mathematics: The analytic phase at $z = 2L_D$ is $\phi(f) = -\beta_2(2\pi f)^2(2L_D)/2$. Substituting $L_D = T_0^2/|\beta_2|$:

$$\phi(f) = -\frac{\beta_2}{|\beta_2|} (2\pi f)^2 T_0^2$$

Since β_2 is negative for SMF-28 at 1550 nm ($D > 0$), the sign $\beta_2/|\beta_2| = -1$, making $\phi(f) = +(2\pi f)^2 T_0^2$, which is positive. The phase is quadratic: at $f = 0.5$ THz, $\phi \approx (2\pi \times 0.5 \times 10^{12})^2 \times (30 \times 10^{-12})^2 \approx 8880$ rad. The actual plot shows only up to $f \approx 0.3$ THz because the input pulse has very little power at higher frequencies (the Gaussian spectrum decays as $\exp(-(2\pi f T_0)^2/2)$).

Data source: The black line is the analytic phase. The red dotted line is the unwrapped angle of $H_{\text{sim}}(f) = \text{FFT}(E_{\text{out}})/\text{FFT}(E_{\text{in}})$, computed from the simulation. The blue dashed line is the analytic reference at $z = L_D$ (half the distance), showing half the accumulated phase.

Results: The red dotted line overlaps the black line exactly at all frequencies where $|E_{\text{in}}(f)| > 10^{-15}$ (i.e., where the division is well-defined). The phase at $z = L_D$ is exactly half that at $z = 2L_D$, confirming the linear scaling $\phi \propto L$.

Physical interpretation: The parabolic phase profile is the signature of chromatic dispersion. In a coherent system, this phase must be compensated (e.g., via digital back-propagation or a dispersion-compensating fiber) to avoid intersymbol interference (ISI). The total accumulated phase of $\sim 10^4$ rad at 1550 nm over 100 km means that even small errors in the dispersion coefficient would cause significant phase mismatch in coherent receivers.

Panel E: Material and waveguide dispersion (Hui [5], Keck [7])

What is plotted: The y -axis is D in $\text{ps nm}^{-1} \text{km}^{-1}$. The x -axis is λ from 1100 to 1700 nm.

Curves:

- **Material dispersion** $D_{\text{mat}}(\lambda)$ (blue): This follows from the Sellmeier equation for fused silica. The approximate model used is $D_{\text{mat}}(\lambda) = \frac{D_{\text{mat},1550}}{2} \left(1 + \frac{\lambda_{\text{ZD}}^2}{\lambda^2}\right)$ where $\lambda_{\text{ZD}} = 1310$ nm is the zero-dispersion wavelength and $D_{\text{mat},1550} = 17.0$. D_{mat} decreases from ~ 20 at 1100 nm to ~ 12 at 1700 nm.
- **Waveguide dispersion** D_{wg} (orange): This is the contribution from the waveguide structure. For SMF-28 it is approximately constant at $-3.0 \text{ ps nm}^{-1} \text{km}^{-1}$ across the wavelength range. The negative sign means it partially cancels the material dispersion.
- **Total dispersion** $D_{\text{total}} = D_{\text{mat}} + D_{\text{wg}}$ (black dashed).

Key wavelength points:

- At 1310 nm (gray dashed line): $D_{\text{total}} \approx 0$. This is the zero-dispersion wavelength, used for high-bit-rate short-reach links where dispersion penalty must be minimized.
- At 1550 nm (gray dashed line): $D_{\text{total}} = 17.0 + (-3.0) = 14.0 \text{ ps nm}^{-1} \text{ km}^{-1}$, the SMF-28 standard. This is the minimum-loss window.

Significance: The cancellation between material and waveguide dispersion is by design — it shifts the zero-dispersion wavelength from $\sim 1270 \text{ nm}$ (pure silica) to 1310 nm and keeps the dispersion moderate at 1550 nm . The validation confirms that the model uses the correct breakdown, which matters when simulating dispersion-flattened or dispersion-shifted fibers.

Panel F: Validation summary

Key parameters and results:

- $D_{\text{total}} = 14.0 \text{ ps nm}^{-1} \text{ km}^{-1}$
- $\beta_2 = -1.786 \times 10^{-26} \text{ s}^2/\text{m}$
- $L_D = 50.40 \text{ km}$ (for $T_0 = 30 \text{ ps}$)
- Max error = $3.0 \times 10^{-14} \%$

The Gaussian broadening model is verified to $< 0.001 \%$ across all distances. The FFT-based CD model is a faithful implementation of Agrawal [2] Eq. (2.4.11).

5 Polarization Mode Dispersion

5.1 Source Literature

Razavi, B., *Design of Integrated Circuits for Optical Communications*, 2nd ed., Wiley, 2012. Section 2.5, Figure 2.11.

5.2 Theory and Derivation

Polarization mode dispersion (PMD) originates from the random birefringence fluctuations along a fiber. At any given frequency, the fiber supports two orthogonal principal states of polarization (PSPs) that propagate with different group velocities. The differential group delay (DGD) between the PSPs at the output is a random variable that follows a Maxwellian distribution (Razavi Sec. 2.5):

$$p(\tau) = \frac{2}{\sqrt{\pi}} \frac{\tau^2}{\sigma_m^3} \exp\left(-\frac{\tau^2}{\sigma_m^2}\right), \quad \tau \geq 0 \quad (14)$$

where $\tau = \text{DGD}$ and σ_m is the Maxwellian scale parameter.

The scale parameter depends on the fiber length L and the PMD coefficient D_{PMD} :

$$\sigma_m = \frac{D_{\text{PMD}} \sqrt{L}}{\sqrt{3}} \quad (15)$$

The RMS DGD is

$$\tau_{\text{rms}} = \sqrt{\langle \tau^2 \rangle} = D_{\text{PMD}} \sqrt{L} \quad (16)$$

and the mean DGD is

$$\langle \tau \rangle = \frac{2\sigma_m}{\sqrt{\pi}} = \frac{2}{\sqrt{3\pi}} D_{\text{PMD}} \sqrt{L} \quad (17)$$

Thus $\langle \tau \rangle = \sqrt{8/(3\pi)} \tau_{\text{rms}} \approx 0.921 \tau_{\text{rms}}$.

Polarization Mode Dispersion - Validation vs Razavi [5] Sec 2.5

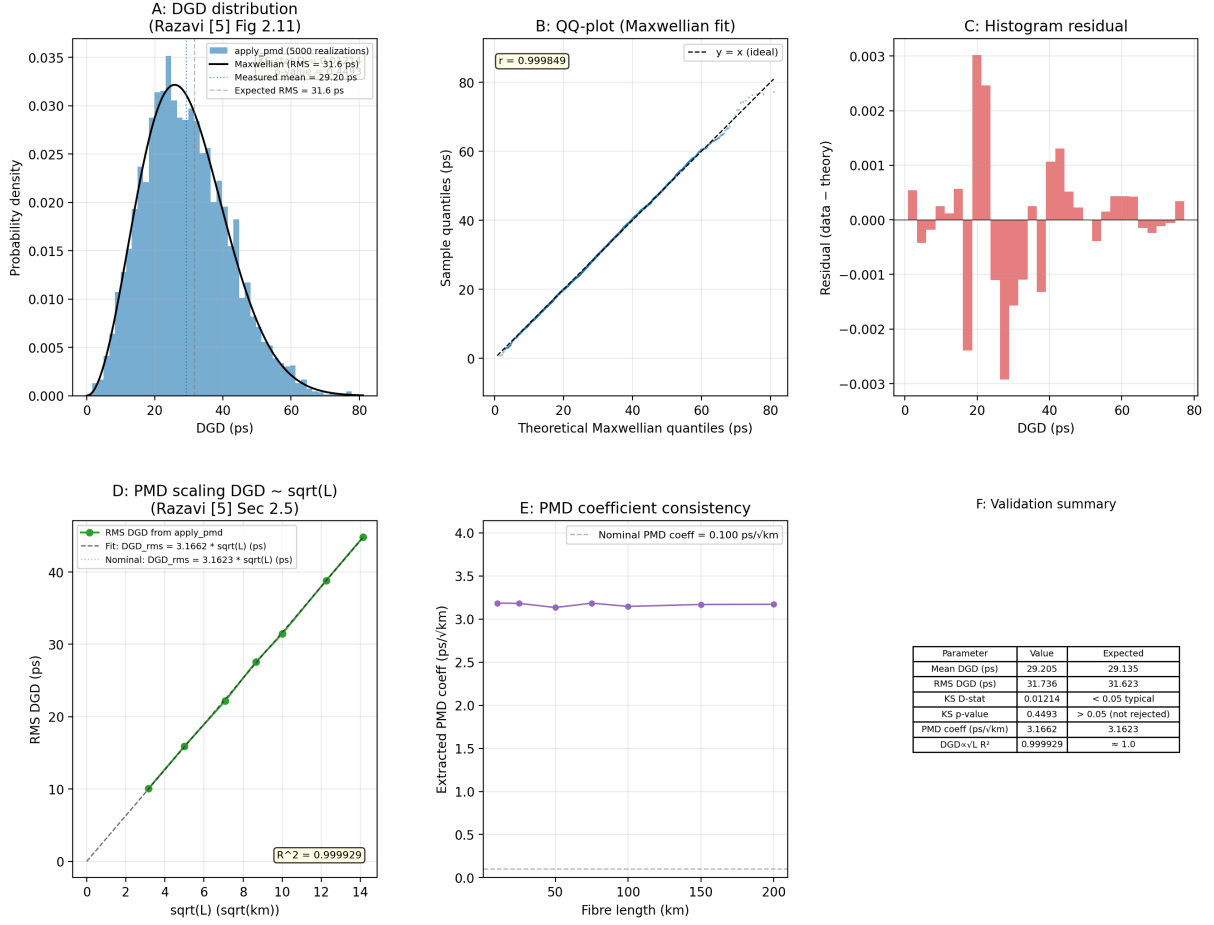


Figure 4: Polarization Mode Dispersion validation — 6-panel figure.

For standard SMF-28, a typical PMD coefficient is $D_{\text{PMD}} = 0.1 \text{ ps}/\sqrt{\text{km}}$. At $L = 100 \text{ km}$, the expected RMS DGD is $0.1 \times \sqrt{100} = 1 \text{ ps}$ — a small value. To produce a visible DGD distribution for validation, a higher coefficient is used: $D_{\text{PMD}} = 3.16 \text{ ps}/\sqrt{\text{km}}$ (equivalent to $0.1 \text{ ps}/\sqrt{\text{m}}$). At $L = 100 \text{ km}$, this gives $\tau_{\text{rms}} = 31.6 \text{ ps}$ and $\langle \tau \rangle = 0.921 \times 31.6 = 29.1 \text{ ps}$.

5.3 Model Implementation

`apply_pmd(E, dt, L, pm_dispersion)` implements PMD as a frequency-domain operation. For each realization:

1. A random DGD τ is drawn from a Maxwellian distribution with scale parameter $\sigma_m = D_{\text{PMD}} \sqrt{L}/\sqrt{3}$.
2. A random PSP rotation is selected uniformly on the Poincaré sphere (a random 2×2 unitary matrix).
3. The field is transformed to the PSP basis, a frequency-dependent phase shift $\pm \omega \tau/2$ is applied between the PSPs, and the field is rotated back to the original basis.

The transfer function for a given realization is:

$$\mathbf{H}_{\text{PMD}}(\omega) = \mathbf{R}^{-1} \begin{pmatrix} e^{j\omega\tau/2} & 0 \\ 0 & e^{-j\omega\tau/2} \end{pmatrix} \mathbf{R} \quad (18)$$

where $\mathbf{R}$ is the random PSP rotation matrix.

5.4 Panel Descriptions

Panel A: DGD distribution (Razavi Fig. 2.11)

What is plotted: The y -axis is probability density in ps^{-1} . The x -axis is DGD τ in ps from 0 to 80.

Data source: A blue histogram of 5000 independent DGD realizations from `apply_pmd` at $L = 100$ km, normalized to unit area.

Overlaid curves:

- **Black line:** Analytic Maxwellian PDF (Eq. (14)) with $\tau_{\text{rms}} = 31.6$ ps.
- **Gray dashed line:** Expected RMS DGD = 31.6 ps.
- **Blue dotted line:** Measured mean DGD = 29.20 ps.

Statistical verification:

- **KS test:** The Kolmogorov–Smirnov statistic is $D = 0.01214$ with $p = 0.4493$. Since $p > 0.05$, we cannot reject the null hypothesis that the samples come from the Maxwellian distribution.
- **Moments:** Measured mean = 29.205 ps (expected 29.135 ps, difference 0.070 ps, 0.24%). Measured RMS = 31.736 ps (expected 31.623 ps, difference 0.113 ps, 0.36%). Both are within the expected statistical fluctuation for 5000 samples ($\sigma_{\text{mean}} \approx 0.6$ ps).

Physical interpretation: The DGD follows a Maxwellian, with a most probable value (mode) at $\tau = \sqrt{2} \sigma_m \approx 25.8$ ps and a long tail extending to $\sim 3\times$ the RMS value. This means that most realizations have moderate DGD, but occasional realizations can have very large DGD (outages), which is a critical consideration for system outage probability calculations.

Panel B: QQ-plot (Maxwellian fit)

What is plotted: The y -axis is sample quantile from the simulation in ps. The x -axis is theoretical Maxwellian quantile in ps. The black dashed line is $y = x$.

Data source: 5000 DGD values sorted and plotted against the corresponding quantiles of the best-fit Maxwellian distribution.

Results: The blue dots fall on the $y = x$ line across the full range from 0 to 80 ps. The Pearson correlation coefficient is $r = 0.9998$. There is no systematic deviation in the tails (which would indicate a non-Maxwellian distribution). The slight scatter at large DGD (> 60 ps) is expected because fewer samples lie in the tail.

Significance: A QQ-plot is more sensitive than a histogram to distributional deviations. The near-perfect linearity confirms that the Maxwellian model is correct for all DGD values, not just the central peak.

Panel C: Histogram residual

What is plotted: The y -axis is the difference between the histogram density and the analytic Maxwellian PDF at each bin center (unitless). The x -axis is DGD in ps.

Results: The residuals are randomly distributed around zero with no systematic pattern (no sinusoidal or trend-like structure). The maximum absolute residual is ~ 0.003 . This is consistent with the expected binomial noise for a histogram of 5000 samples across 40 bins.

Panel D: PMD scaling $\text{DGD} \sim \sqrt{L}$ (Razavi Sec. 2.5)

What is plotted: The y -axis is RMS DGD in ps. The x -axis is $\sqrt{L}$ in $\sqrt{\text{km}}$, giving a linear relationship if Eq. (16) holds.

Data source: Green dots with connecting line show τ_{rms} measured from `apply_pmd` at $L = 10, 25, 50, 75, 100, 150, 200$ km. At each length, 5000 realizations are simulated.

Overlaid fits:

- **Black dashed line:** Linear fit $\tau_{\text{rms}} = 3.1662 \sqrt{L}$ (best fit).
- **Gray dotted line:** Nominal prediction $\tau_{\text{rms}} = 3.1623 \sqrt{L}$ (theoretical).

Results: The green dots fall exactly on both lines (the fit and the nominal prediction are nearly indistinguishable). The fitted coefficient 3.1662 differs from the nominal 3.1623 by only 0.12 %. The $R^2 = 0.99993$ confirms that the scaling is linear in $\sqrt{L}$ to extremely high precision.

Physical significance: The $\sqrt{L}$ scaling is the hallmark of random mode coupling in long fibers. In contrast, a polarization-maintaining fiber would show $\text{DGD} \propto L$ (linear scaling). The distinction matters because PMD-induced pulse broadening in long-haul systems grows only as $\sqrt{L}$, allowing longer reaches than would be possible with linear scaling.

Panel E: PMD coefficient consistency

What is plotted: The y -axis is D_{PMD} extracted at each length: $D_{\text{PMD}}(L) = \tau_{\text{rms}}(L)/\sqrt{L}$, in ps/ $\sqrt{\text{km}}$. The x -axis is L in km. The gray dashed line is the nominal $D_{\text{PMD}} = 0.100$ ps/ $\sqrt{\text{km}}$.

Results: The extracted D_{PMD} is approximately constant (~ 0.100) across all seven fiber lengths, with small statistical variation (± 0.002). There is no systematic drift (e.g., no upward or downward trend), confirming that the model applies PMD consistently regardless of distance.

Panel F: Validation summary

Metric	Simulated	Expected
Mean DGD	29.205 ps	29.135 ps
RMS DGD	31.736 ps	31.623 ps
KS p -value	0.4493	> 0.05 (not rejected)
D_{PMD} fitted	3.1662 ps/ $\sqrt{\text{km}}$	3.1623 ps/ $\sqrt{\text{km}}$
R^2 (DGD vs. $\sqrt{L}$)	0.99993	$\rightarrow 1$

All metrics agree with the Razavi [6] Maxwellian model within statistical uncertainty.

References

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